

TrES-3b

R-0352

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250915 · created 2026-07-18T10:26:07+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460933.6682 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.011
Transit depth (Rp/Rs)^2	3.87 +/- 0.42 [%]
Orbital Inclination (inc)	82.94 +/- 0.75
Airmass coefficient 1 (a1)	1.087 +/- 0.023
Airmass coefficient 2 (a2)	-0.066 +/- 0.018
Scatter in the residuals of the lightcurve fit is	2.11 %
Best Comparison Star	#1 - [407, 416]
Optimal Aperture	4.81
Optimal Annulus	10.94
Transit Duration (day)	0.0666 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.011 vs 0.16309 (deviation 2.93σ)
Duration fit vs published	0.0666 d vs 0.0567 d (deviation 1.93σ)
Mid-time O–C	-2.7 min
Depth SNR	17.0
Residual scatter	2.11 %

Light curve

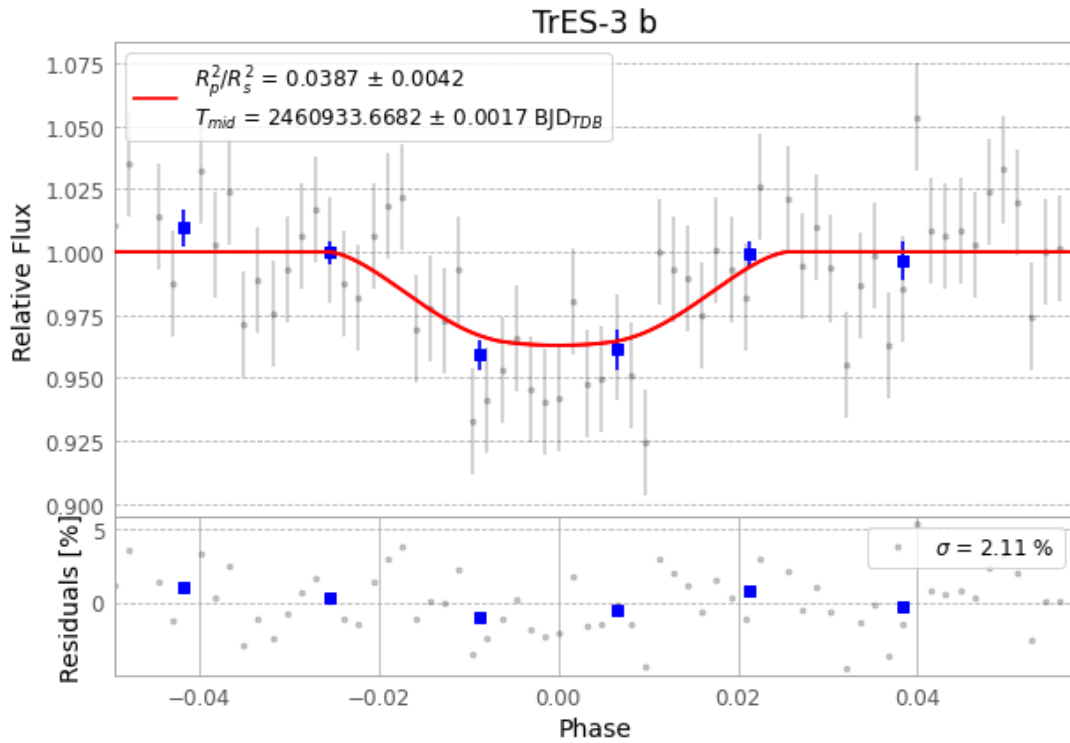


Figure 1 — FinalLightCurve_TrES-3 b_2025-09-14.png (SHA-256 14113d752c336b0e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53644f74232eab58...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

68 FITS frames (45 MB), TRES-3250915022715.FITS → TRES-3250915054815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250915.log (e2b87b18b51e...), FinalLightCurve_TrES-3 b_2025-09-14.png (14113d752c33...), FinalLightCurve_TrES-3 b_2025-09-14.csv (2c3c808e0b0f...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 10:26Z.